Date

Expt. No. 1 Page No. 1

Aim of the Experiment

To identify various types of cables & connectors used in computer or tele-Network.

Apparatus Required

- Twisted Pair cables (Cat-3, Cat-5e, Cat-6/6a, Cat-8)
- Fiber Optic cables (SMF, MMF)
- Coaxial cables (RG6, RG11, RG58, RG59)
- Audio/video and peripheral cables (HDMI, USB, SATA, VGA, etc.)
- Corresponding connectors for all cable types.

Theory

→ What are cables & connectors in a Computer Network

In a computer Network, cables serve as the physical transmission medium that carries data signals from one network device to another. Connectors are the specialized electromechanical commponents attached to the end of these cables, allowing them to interface securely with the ports on network hardware like computers, switches and routers.

→ Why are they required?

They are fundamentally required to establish the physical layer of a wired network. without

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devices can't reliably transmit the electrical or optical signals needed for ~~comm~~ comn. They provide a dedicated secure pathway for data guiding the signal accurately to the destination.

Types of cables

1. Twisted pair cables

- This type of cable features pairs of insulated copper wires twisted together, a design that effectively cancels out electromagnetic interfa (EMI) and crosstalk from adjacent pairs.

- The most common connectors used for these are the RJ45, RJ11, RJ48 etc.

2. Fiber Optics Cables

- Fiber optic cable transmit data using rapidly pulsing beams of light through highly transparent microscopic glass or plastic cores, offering immense bandwidth and total immunity to electrical interf. They are classified into single Mode fiber (SMF), Multimode Fiber (MMF)

- Precision control align the glass core to prevent light loss, including LC, SC, ST, MPO/MTP for high density appln.

Date

Expt. No. 1 Page No. 3

3. Coaxial cables

-> It consist of a central copper conductor surrounded by a dielectric insulating layer, a metalic shield (braid or foil), and an outer plastic jacket to prevent signal leakage and external interference. They categorise by radioguide (RG), RG6, RG11, RG58, RG59 etc.

Conclusion

A comprehensive range of networking transmission media and ~~other~~ peripheral connections were successfully identified and studied.

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Expt. No. 2 Page No. 4

Aim of the Exp

To study the detailed technical specifications, characterists, and performance metrics of various computer network cables and their respective connectors.

Apparatus Required.

- Detailed diagrams are color prints of networking cables and connectors.
- Specification data sheets for standard network media.
- Samples of Twisted pair (Cat 5e/Cat 6), Coaxial (RG-6/RG59) and fiber optic (SMF/MMF) cables (if available for refrence)

Theory

1. Twisted Pair Cable
Twisted Pair Cable is the most commonly used cable in LAN (Local Area Network). It consists of pairs of insulated copper wires twisted together to reduce electromagnetic interference.

(*) Types of Twisted pair cable, (i) UTP (unshielded Twisted Pair)
(ii) STP (Shielded Twisted Pair)

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(1) UTP → (Unshielded Twisted Pair)
→ It Does not contain additional shielding.

→ It's widely used in Ethernet networks.

→ Cheaper and very easy to install.

(2) STP (Shielded Twisted Pair)

→ Contains metal shielding to reduce interference

→ Used in environments with high Electromagnetic noise

★ Categories of Twisted pair cable (Specifications)

Category	Maximum Speed	Bandwidth	Uses
Cat 5	100 Mbps	100MHz	Fast Ethernet
Cat 5e	1 Gbps	100MHz	Gigabit Ethernet
Cat 6	10 Gbps (short range)	250MHz	High Speed Networks
Cat 6a	10 Gbps	500MHz	Datacenters
Cat 7	10 Gbps +	600MHz	High performance Networks

★ Connectors used
RJ45: It is an 8-Position-8 contact (8P8C) modular connector. The internal pins are usually goldplated (e.g. 50-micron gold plating) to prevent corrosion and

Expt. No. 2 Page No. 6

Date

ensure max^m conductivity.

→ They are wired according to specific colon-code standards, primary TB68A or T568B.

(★) ~~characterists~~ → Low cost, Max^m length: 100 meters Easy to install.

2. Coaxial Cables

→ These cable is designed to transmit high-frequency electrical signals with low loss. It features a concentric topological design; a central solid copper or copper clad steel core conductor, surrounded by an insulating dielectric foam or solid Teflon.

→ This is encased in tubular conducting shield (usually a woven copper braid) that serves as the ground and protects against EMI, all wrapped in a pvc conductor.

(★) Types of Coaxial Cable

① Thick Coaxial Cable (10 Base 5)

→ used in early Ethernet networks

→ Longer distance support

② Thin Coaxial Cable (10 Base 2)

→ Thinner and more flexible

→ used in small networks.

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Expt. No. 2 Page No. 7

Categorization

Category	Ethernet standard	Max^n length	connector
Thick coaxial	10Base5	500m	N-type
Thin coaxial	10Base 2	185m	BNC

Connector

→ BNC connector → It features a bayonet-style, twist and lock mechanism ensuring a tight, secure connection that won't accidentally pull out, commonly ~~use~~ rated for frequencies upto 4GHz. ~~The F-type connector uses~~

→ The N-type connector uses a threaded screw on mechanism and ~~ingeniously utilizes the solid central conductor of the coaxial cable itself as the connector central pin, minimizing signal loss for broadband applications.~~

Characterists

- Better shielding than twisted pair
- Supports longer distances
- Expensive than twisted pair
- Difficult to install compared to UTP.

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Date

Expt. No. 2 Page No. 8

3. Fiber optics cables

→ Fiber optics represents a fundamental shift from electrical signaling to optical (light) signaling. The cable consists of an ultra-pure glass or plastic core surrounded by a cladding layer with a slightly lower refractive Index. This refractive Index causes "TIR", trapping the light pulses inside the core as they travel.

Ⓐ Types of fiber optics

1// Single Mode Fiber (SMF)
→ It's have very small core diameter
→ It's used for long-distance communication
→ used in telecom networks.

2// Multimode Fiber (MMF)
→ Large core diameter

→ used for shorter distance such as LAN.

Connectors used

(I) ST (Straight Tip) (III) LC (Lucent connector)

(II) SC (Subscriber Connector). (IV) FC (Ferrule Connector)

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(4) Characterists
→ very high bandwidth.
→ long-distance transmission.
→ Immune to Electromagnetic interference.
→ High installation cost.
→ Fragile compared to copper cables.

⇒ Performance Metrics of Network cables
The performance of network cable is measured using several parameters:
1. Bandwidth → The max^m data transfer capacity of cable.
2. Data Transmission speed → rate at which data can be transmitted.
3. Attenuation → Loss of signal strength over distance.
4. Noise/Interference → Disturbance affecting signal Quality
5. Max^m Distance → The maximum length over which Data can be transmitted without signal loss.
6. Reliability → The ability to maintain stable comm^n without errors.

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Conclusion :- The detailed specifications, physical characteristics and performance - metrix of standard network cables and connectors were successfully studied.

12.02.26

Exp-3

Aim of the Experiment: Making patch cords using different types of cables and connectors - crimping, splicing, etc.

Apparatus Required

1/ Cables: Cat 5e, Cat 6, Fiber optic, Coaxial
2/ Connectors: RJ45, BNC, SC, LC.
3/ Tools: Crimping tool, cable stripper, cable cutter, splicing kit.
4/ Consumables: Heat shrink tubing (optional).
5/ Diagnostic: Network/Cable tester.

Theory

① Crimping Tool: A specialized hand tool used to join a connector (like RJ45) to the end of a wire by deforming the metal part of the connector to hold the wire securely.

② Cable stripper: Designed to remove the outer insulation or protective sheath from electrical or optical cables without damaging the internal conductors.

③ Cable cutter: Uses a scissor-like shearing action to cut thick cables cleanly, preventing the 'crushing' effect caused by standard pliers.

Date			
Page No.	2		

3

$) **Splicing kit:** A set used to permanently join two or more cables. Unlike crimping (which is for terminals), splicing is used for permanent repair or extensions.

$) **Heat shrink Tubing:** A shrinkable plastic tube used to insulate joints, providing abrasion resistance and environmental protection.

$) **Tester:** A diagnostic tool used to determine the electrical signal, voltage, or continuity in a completed circuit.

procedure: making patch cords

1/ Ethernet patch cords (Cat 5e/6 with RJ45)

-> Ethernet patch cords are flexible network cable used to connect devices (like a pc to a router) for signal routing.

Steps

1/ Cut the Cable: Cut the Ethernet cable to the desired length using a cable cutter.

2/ Strip the Jacket: Use a stripper to remove approximately 1 inch of the outer Jacket.

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3// Untwist $ Align: Untwist the wire pairs and align them according to the T568A or T568B wiring standard.

4// Trim: Trim the wires evenly so they sit flush inside the connector.

5// Insert: Push the wires into the RJ45 connector, ensuring each wire reaches the end.

6// Crimp: place the connector into the crimping tool and squeeze firmly to secure the connection.

7// Test: use a network cable tester to ensure there are no faults or crossed wires

2// Coaxial patch cords (with BNC connectors)

Coaxial cables consist of single copper conductor surrounded by layers of insulation and shieldings used for high-frequency signals with low loss.

Steps:

1// Cut: Cut the coaxial cable to the required length.

2// Strip: strip the outer jacket, shield, and dielectric to expose the center conductor.

3// Prepare connectors: Slide the crimpsleeve onto the cable before attaching the head.

No. 3

Date | Page No. 4

4// **Attach**: Insert the cable into the BNC connector, ensuring the center conductor enters the center pin.

5// **Crimp**: Use the crimping tool to crimp the sleeve onto the connector body.

6// **Test**: check for signal continuity using a coaxial cable tester.

3// **Fiber optic Patch cords (with SC/LC connectors)**
Fiber cords use pulses of light to transmit data through thin strands of glass or plastic.

Steps:

1// Cut: cut the fiber optic to length.
2// Strip: Use a specialized fiber stripper to remove the outer jacket and buffer coating.
3// Cleave: Use a fiber cleaver to create a clean, perpendicular cut on the fiber end.
4// Prepare: slide the crimp sleeve and protective boot onto the fiber.
5// insert: Insert the fiber into the connector; use a splicing tool if the connector requires it.

6// Crimp: secure the connector by crimping the sleeve.

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3

7/ polish: polish the fiber end using a fine polishing film to ensure a smooth surface for light transmission.

8/ Test: Use an optical power meter or visual fault locator (VFL) to verify the connection.

conclusion

The patch cord of cables and connectors crimping splicing is successfully made in this experiment.

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Date			
Page No.	1		

4

Aim of the Experiment:

To Demonstratation of diff^n type of cable testers using them for testing patch cords prepared by the students in Lab and standard cables prepared by professionals.

Apparatus Required

(i) Basic Ethernet cable
(ii) Advanced Network cable tester
(iii) Basic Coaxial Cable Tester
(iv) Visual fault ~~Det~~ locator (VFL)
(v) optical Power meter

Theory

① Ethernet cable Tester: A diagnostic tool used to verify electrical connections and physical integrity of networking cables like Cat5e, cat6, or cat 6a etc. It Ensures signals travel from one end to the other without interruption.

② Advanced Network cable Tester: often categorized as an Analyzer or Certifier. It is a high-precision instrument that verifies if a cable meets international performance standards (e.g., ISO/IEC 11801). Unlike basic testers, it ensures the cable can support high speed data rates like 10 Gbps by analyzing the physical layer of the OSI Model

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Date
Page No.

Procedure

- Using a Basic Tester:

1. Connect one end of the Ethernet cable to the main unit and the other to the remote unit.

2. Turn on the tester and observe the LED indicators.

3. Interpretation: Green lights indicate correct wiring, while,

- Using an Advanced Tester:

1. Connect the cable: plug the cable into the high-precision tester.

2. Select Test Mode: Choose the appropriate mode, such as continuity, wire map, or length.

3. Start the Test: Begin the automated test and observe the detailed results on the digital display.

(1) Basic Ethernet cable Tester: For verifying electrical connections and physical integrity.

(2) Advanced Network Cable Tester: A high-precision diagnostic instrumental/analyzer.

(3) Basic coaxial cable Tester: For checking coaxial cable.

Teacher's Signature........................

Date
Page No.

(A) Visual Fault Locator (VFL): For fiber-optic fault detection.

(A) Optical power Meter: For measuring fiber optic signal strength make these few longer.

→ Demonstration & comparison.

→ Test a student-prepared cable and record the results.

→ Test a professional prepared cable and compare the results.

Conclusion:

we have succesfully demonstrated the ~~Demonstration of~~ diffⁿ types of cable testers using them for testing patch cords.

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No. 5

Date
Page No.

Aim of the Exp.

~~Con~~ To configure computing devices (PC, Laptop, Mobile, etc) for network, exploring diffn options and their impact - IP address, gateway, DNS, security options, etc.

Apparatus Required

> Network Interface cards (NICs).

> Hubs and switches

> Routers and wifi Access points (AP).

Theory & Functions

1 Network Interface Card (NIC): A hardware component also known as a network.

• Ethernet NIC: Used for wired connections typically via RJ45 connectors.

• Wireless NIC: Equipped with Wi-fi capabilities for wireless connectivity.

• Usage: provides the physical interface for communication; can be internal or connected USB.

Teacher's Signature.........................

Date | Page No.

2. Hub: A basic Layer 1 (physical layer) devices that connects multiple Ethernet devices into a single network segment.

→ passive Hubs: simply connect devices without amplifying the signal.

→ Active Hubs: Amplify the incoming signal before broadcasting it to all ports.

→ Limitation: Not commonly used in modern networks due to inefficiency and data collision issues.

3. Switch: A layer 2 (Data link layer) device that ~~connector~~ connects devices and uses MAC Adresses to forward data only to the intended recipient.

→ Unmanaged Switch: provide basic functinality without configuration option

→ Managed switch: supports advanced features like VLANs and Quality sevice.

4. Router → A router acts as a central hub that connects multiple
~~4. WiFi Access Point~~ networks to the internet, Ex: house Router.

Types

(i) wired Router.

(ii) wireless Router

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Date

Expt. No. Page No.

5. Wifi Access point (AP):

→ Acts as bridge between wireless devices and a wired Network.

→ standalone AP: Managed individually

→ controller based AP: Managed centrally through a wireless controller.

Conclusion:

we have successfully configured computing devices (PC, Laptop, mobile, etc) for Network, exploring diffⁿ options and their impact - IP adress, gateway, DNS, security options etc.

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